

# PE-EC703A — Embedded System

Semester VII • 3L:0T:0P • 3 Credits

## Module 5 Exam Notes: Embedded System Design using PIC Microcontroller

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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## 1. Introduction to Microchip PIC16 Family

### PIC Microcontroller

**PIC** (Peripheral Interface Controller) is a family of microcontrollers manufactured by **Microchip Technology**. PIC microcontrollers are known for their RISC-based Harvard architecture, a small instruction set (<35 instructions for mid-range PIC), low cost, wide voltage range (2.0–5.5 V), and abundant peripheral integration. They are widely used in hobbyist and industrial embedded applications.

### 1.1 PIC Architecture Classifications

| Family            | Program Word | Data Bus | Examples                         |
|-------------------|--------------|----------|----------------------------------|
| PIC10 (Base)      | 12-bit       | 8-bit    | PIC10F200, PIC10F220             |
| PIC12 (Base)      | 12-bit       | 8-bit    | PIC12F508, PIC12F675             |
| PIC16 (Mid-range) | 14-bit       | 8-bit    | PIC16F84A, PIC16F877A, PIC16F873 |
| PIC18 (Enhanced)  | 16-bit       | 8-bit    | PIC18F4550, PIC18F452            |
| PIC24 / dsPIC     | 24-bit       | 16-bit   | dsPIC33, PIC24FJ                 |
| PIC32             | 32-bit MIPS  | 32-bit   | PIC32MX, PIC32MZ                 |

### 1.2 Key Architectural Features of PIC Mid-Range (PIC16)

- **Harvard Architecture:** Separate program (Flash) and data (RAM + SFR) buses.
- **RISC Core:** Only 35 instructions; most execute in 1 instruction cycle (4 clock cycles).
- **Instruction Cycle:**  $T_{cy} = 4/f_{OSC}$  (e.g., at 20 MHz:  $T_{cy} = 200$  ns).
- **8-bit Working Register (W):** Equivalent to the Accumulator in 8051.
- **File Register Architecture:** All RAM, SFRs, and I/O registers accessed via “file register” addresses.
- **Pipelining:** 2-stage pipeline (Fetch + Execute) — while one instruction executes, the next is fetched.

### PIC16 Key Formulas

$$T_{cy} = \frac{4}{f_{OSC}} \quad (1 \text{ instruction cycle} = 4 \text{ clock periods})$$

$$\text{Timer0 Overflow Time} = \frac{(256 - \text{TMR0}) \times \text{Prescaler} \times 4}{f_{OSC}}$$

$$\text{ADC Resolution} = \frac{V_{REF}}{2^N - 1} \quad (N = 10 \text{ bits for PIC16F873})$$

$$\text{Baud Rate (UART)} = \frac{f_{osc}}{64 \times (\text{SPBRG} + 1)} \quad (\text{Low speed mode})$$

## 2. PIC16F873 — Architecture and Features

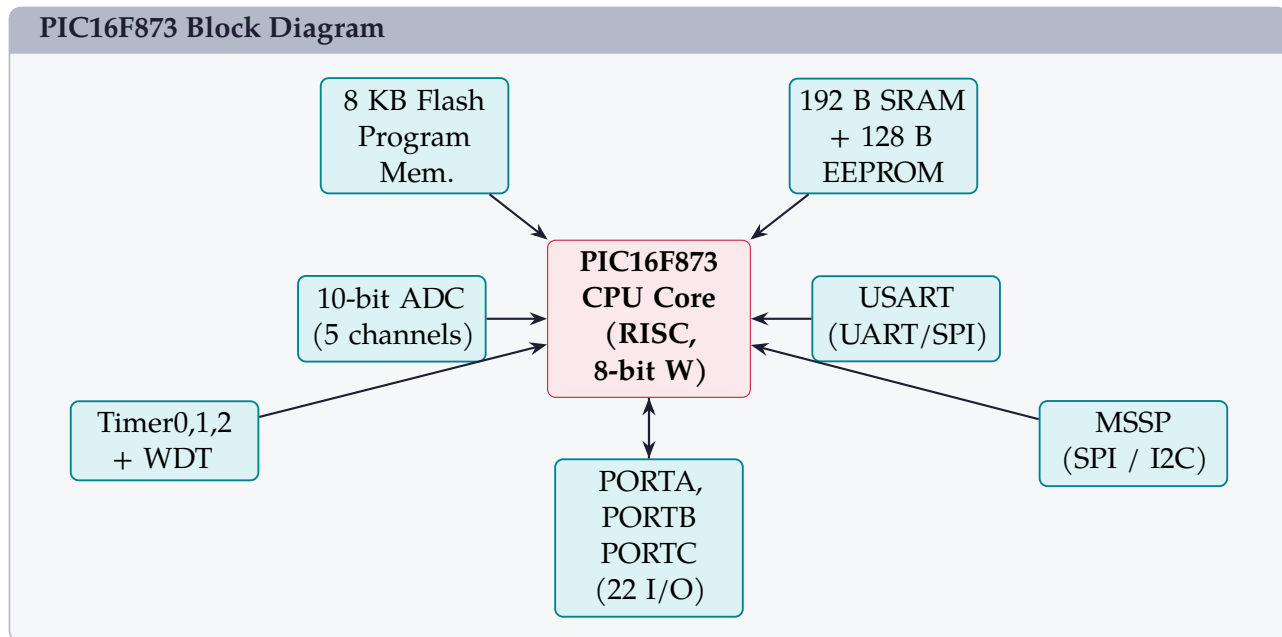
### PIC16F873

The **PIC16F873** is a 28-pin Flash-based mid-range PIC microcontroller from Microchip. It is a popular choice for educational and industrial embedded designs due to its integrated analog and communication peripherals, Flash programmability, and low power consumption.

### 2.1 PIC16F873 — Key Specifications

| Feature           | Specification                                           |
|-------------------|---------------------------------------------------------|
| Program Memory    | 8 KB Flash (4096 x 14-bit words)                        |
| Data Memory (RAM) | 192 bytes SRAM                                          |
| Data EEPROM       | 128 bytes                                               |
| I/O Ports         | PORTA (6 pins), PORTB (8 pins), PORTC (8 pins) = 22 I/O |
| Timers            | Timer0 (8-bit), Timer1 (16-bit), Timer2 (8-bit + PR2)   |
| ADC               | 5-channel, 10-bit ADC (AN0–AN4 on PORTA)                |
| Serial Ports      | USART (UART/SPI async), MSSP (SPI + I2C)                |
| Interrupts        | Up to 13 interrupt sources                              |
| Clock Speed       | DC to 20 MHz (using external crystal)                   |
| Supply Voltage    | 2.0 V to 5.5 V                                          |
| Package           | 28-pin PDIP, SOIC, SSOP                                 |
| Watchdog Timer    | Built-in WDT with independent RC oscillator             |

PIC16F873 Block Diagram



### 3. On-Chip Peripherals of PIC16F873

#### 3.1 Watchdog Timer (WDT)

- Has its own internal **RC oscillator** (independent of main clock); continues running even if main clock fails.
- Default timeout period  $\approx 18$  ms; extended via **postscaler** (up to  $\times 128 \approx 2.3$  s).
- Reset by `CLRWDT` instruction in software.
- On timeout: generates a device reset (MCLR pin goes low momentarily).
- Can be enabled/disabled via configuration word bits.

#### 3.2 ADC (Analog-to-Digital Converter)

- **10-bit successive approximation ADC**: Converts analog voltage to a 10-bit digital value (0 to 1023).
- **5 analog input channels** (AN0–AN4) on PORTA (multiplexed with digital I/O).
- **Reference voltage**:  $V_{REF+}$  and  $V_{REF-}$  pins (or  $V_{DD}$  and  $V_{SS}$  by default).
- **Acquisition time**: Capacitor sampling time before conversion starts; minimum  $2 \mu\text{s}$ .
- **Result format**: Left-justified or right-justified; stored in ADRESH:ADRESL registers.

#### PIC16F873 ADC — Conversion Steps

1. Configure ADCON1 (select analog/digital pin assignment and  $V_{REF}$ ).
2. Configure ADCON0 (select channel AN $x$ , clock source, turn ADC ON).
3. Wait for acquisition time ( $\geq 2 \mu\text{s}$ ).
4. Set GO/DONE bit in ADCON0 to start conversion.

5. Poll `GO/DONE` bit until it clears (conversion complete,  $\approx 12 \times T_{AD}$ ).
6. Read 10-bit result from `ADRESH` (high 2 bits) and `ADRESL` (low 8 bits).

### 3.3 Asynchronous Serial Port — USART

- Full-duplex UART on RC6 (TX) and RC7 (RX).
- Baud rate controlled by SPBRG register and BRGH bit (high/low speed).
- **High-speed mode (BRGH=1):**  $\text{Baud} = f_{OSC} / (16 \times (\text{SPBRG} + 1))$
- **Low-speed mode (BRGH=0):**  $\text{Baud} = f_{OSC} / (64 \times (\text{SPBRG} + 1))$
- Interrupt flags: TXIF (transmit buffer empty), RCIF (receive buffer full).

### 3.4 SPI and I2C — MSSP Module

- The **MSSP** (Master Synchronous Serial Port) supports both SPI and I2C modes.
- **SPI Mode:** Uses RC3 (SCK), RC4 (SDI), RC5 (SDO), and any GPIO as CS (Chip Select). Master or slave configurable.
- **I2C Mode:** Uses RC3 (SCL) and RC4 (SDA) with open-drain outputs. Supports 7-bit and 10-bit addressing.

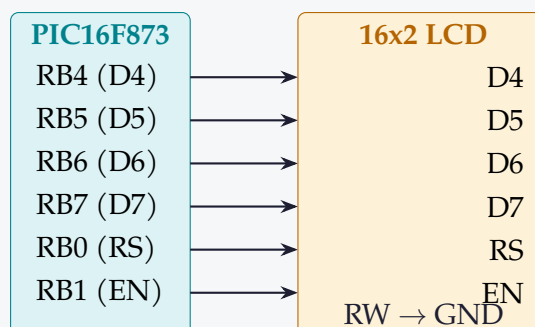
## 4. Interfacing with PIC16F873

### 4.1 LCD Interfacing

#### LCD (16×2) Interfacing with PIC16F873

A standard Hitachi HD44780-compatible 16×2 character LCD is interfaced using **4-bit mode** (4 data lines + 3 control lines) to save I/O pins. Control pins: RS (Register Select: 0=Command, 1=Data), RW (Read/Write, tied to GND for write-only), EN (Enable pulse to latch data).

#### PIC16F873 to 16×2 LCD — 4-bit Interface



### 4.2 ADC Interfacing (Sensors)

- Temperature sensor (LM35): Output voltage = 10 mV/°C. Connect to AN0.

– ADC result → Temperature:  $T(^{\circ}C) = \frac{ADC \times V_{REF}}{1023 \times 0.01}$

- Potentiometer / Joystick: Directly connected to analog input (voltage divider).
- Light sensor (LDR): Used in voltage divider; ADC reads light level.

### 4.3 Stepper Motor Interfacing

- A stepper motor advances by a fixed angle (step) for each control pulse.
- **Step angle:** Depends on motor type (e.g.,  $1.8^{\circ}/\text{step} = 200 \text{ steps/revolution}$ ).
- **Driver IC:** L298N or ULN2003 connects between PIC output pins and motor coils.
- PIC sends sequence of bit patterns to PORTB to rotate the stepper.
- **Full-step sequence (4-phase):** 1100 → 0110 → 0011 → 1001 (repeat).

### 4.4 Keyboard (Keypad) Interfacing

- A **4×4 matrix keypad** uses 4 row + 4 column lines = 8 pins.
- PIC drives rows LOW one at a time, then reads which column line goes LOW.
- The row/column combination identifies the pressed key.
- Debouncing: Software delay ( $\sim 20 \text{ ms}$ ) after detecting key press before confirming it.

### 4.5 DAC (Digital-to-Analog Converter)

- PIC16F873 has no built-in DAC. External options:
  - **R-2R Resistor Ladder:** Passive DAC using resistors connected to PORTB (8-bit). Simple, low cost, no extra IC.
  - **External DAC IC (MCP4921):** 12-bit DAC with SPI interface.
  - **PWM + RC filter:** Use Timer2 PWM output, filter with RC low-pass → analog voltage proportional to duty cycle.

## 5. Case Studies of Embedded System Design

### 5.1 Case Study 1: Automated Chocolate Vending Machine

#### Automated Chocolate Vending Machine using PIC16F873

**Problem:** Design a vending machine that accepts coins (detected by IR sensor), displays balance on LCD, dispenses chocolate (stepper motor) when enough coins are inserted, and provides change.

#### Hardware Components:

- PIC16F873 (controller), 16×2 LCD (display), IR coin sensor on AN0 (ADC), Stepper motor + ULN2003 (dispenser), Buzzer on RB0 (feedback), Keypad for item selection.

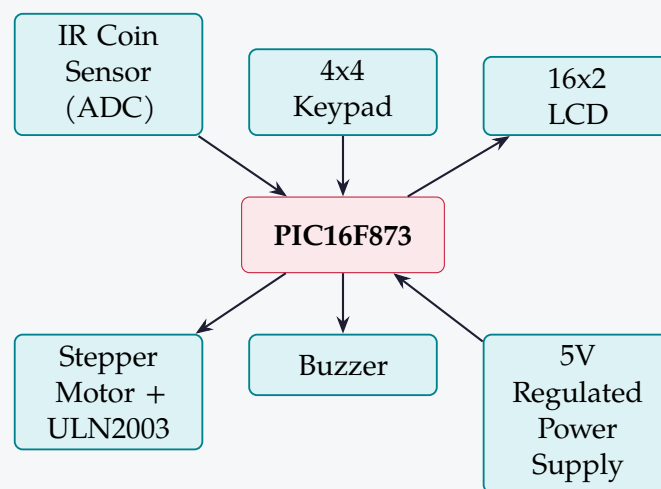
#### Software Flow:

1. Initialize LCD, ADC, I/O ports.

2. Display “Insert Coin” on LCD.
3. Loop: Poll coin sensor (ADC reading > threshold → coin detected).
4. On coin: Increment balance; update LCD display.
5. If balance  $\geq$  price: Drive stepper motor (4 steps) to dispense; beep buzzer.
6. Calculate and dispense change (reverse stepper? No — separate change mechanism).
7. Reset balance; display “Thank You”.

**Key Programming Concepts Used:** ADC polling, stepper motor control sequences, LCD 4-bit mode, software debounce, timer-based delay.

#### Vending Machine System Block Diagram



## 5.2 Case Study 2: Digital Camera Controller

### Digital Camera Controller using PIC (Concept)

**Note:** A full digital camera uses high-performance processors (e.g., ARM Cortex-A) for image processing. PIC16F873 can implement the **camera control subsystem** (not image processing).

#### PIC16F873 Role in Digital Camera:

- **Shutter Control:** Timer-based precise shutter speed control (1/1000s to 1/30s) using Timer1 interrupt.
- **Focus Motor Control:** Stepper motor drives lens position; PIC reads ADC from autofocus sensor.
- **Flash Trigger:** GPIO pin triggers external flash unit via optocoupler.
- **User Interface:** Reads mode dial (keypad), displays settings on small LCD.
- **Battery Monitor:** ADC reads battery voltage divider; displays charge level.
- **Serial Communication:** UART or SPI to main image processor (ARM) for command passing.

**Key Techniques:** Timer interrupts for precision timing, ADC for sensor reading, SPI for inter-processor communication, WDT for reliability.

## Quick Revision — Module 5

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### Key Formulas at a Glance

|                               |                                                                                  |
|-------------------------------|----------------------------------------------------------------------------------|
| <b>Instruction Cycle</b>      | $T_{cy} = \frac{4}{f_{OSC}}$ (at 20 MHz: $T_{cy} = 200$ ns)                      |
| <b>Timer0 Overflow</b>        | $t_{TMR0} = \frac{(256 - TMR0\_init) \times \text{Prescaler} \times 4}{f_{OSC}}$ |
| <b>ADC Result (10-bit)</b>    | $V_{in} = \frac{ADC \times V_{REF}}{1023}$ (right-justified)                     |
| <b>UART Baud (BRGH=1)</b>     | $\text{Baud} = \frac{f_{OSC}}{16 \times (\text{SPBRG} + 1)}$                     |
| <b>ADC Temperature (LM35)</b> | $T(^{\circ}\text{C}) = \frac{ADC \times V_{REF}}{1023 \times 0.01}$              |

### One-Line Definitions

- **PIC:** Peripheral Interface Controller — Microchip Technology's RISC-based MCU family.
- **PIC16F873:** 28-pin, 8-bit Flash PIC MCU; 8 KB Flash, 192 B SRAM, 10-bit ADC, USART, MSSP.
- **W Register:** Working Register in PIC — equivalent to Accumulator; all ALU operations use W.
- **File Register:** PIC's general-purpose RAM + SFR memory, accessed via file register addresses.
- **ADCON0/1:** ADC Control registers in PIC16F873 that configure channel selection, clock, and reference.
- **MSSP:** Master Synchronous Serial Port — supports SPI and I2C in PIC16F873.
- **USART:** Universal Synchronous/Asynchronous Receiver/Transmitter — UART + SPI on PIC.
- **SPBRG:** Serial Port Baud Rate Generator register — sets UART baud rate in PIC.
- **Prescaler:** Divides the clock signal before reaching the timer — extends the timer overflow period.
- **TMR0:** 8-bit Timer/Counter in PIC16F873 (overflows at 255 → 0, generates TMR0IF).
- **CLRWDT:** Assembly instruction to reset the Watchdog Timer in PIC.
- **Full-step sequence:** 4-phase stepper motor control pattern (1100, 0110, 0011, 1001).

## Frequently Asked / Viva Questions

**Q1. What does PIC stand for and who manufactures it?**

PIC stands for **Peripheral Interface Controller**. It is manufactured by **Microchip Technology Inc.**

**Q2. State the key features of PIC16F873.**

28-pin PDIP; 8 KB Flash program memory; 192 B SRAM; 128 B EEPROM; 10-bit 5-channel ADC; three timers (Timer0/1/2); USART; MSSP (SPI + I2C); up to 20 MHz clock; 2.0–5.5 V supply; built-in WDT.

**Q3. What is the W register in PIC? How is it different from 8051's Accumulator?**

The **W (Working) register** is PIC's primary data register used for ALU operations. Similar to the Accumulator in 8051, it is used as the source/destination for most arithmetic and logic instructions. Unlike 8051's A, W is not directly addressable in the file register map.

**Q4. What is the instruction cycle of PIC16 and how does it relate to clock frequency?**

One instruction cycle = 4 clock cycles:  $T_{cy} = 4/f_{OSC}$ . At 20 MHz,  $T_{cy} = 200$  ns. Most PIC instructions execute in 1 cycle (200 ns); branch instructions take 2 cycles.

**Q5. How does the PIC16F873 ADC work? What is the resolution?**

It is a 10-bit successive approximation ADC. It samples the analog voltage on AN0–AN4, converts it to a 10-bit number (0–1023) representing  $V_{in}$  between  $V_{REF-}$  and  $V_{REF+}$ . Resolution =  $V_{REF}/1023$  (e.g.,  $\approx 4.88$  mV for 5V reference).

**Q6. Explain how a stepper motor is interfaced with PIC16F873.**

The PIC outputs a 4-bit step sequence to PORTB (e.g., 1100, 0110, 0011, 1001 for full-step mode). Each pattern energizes the motor coils in sequence. A ULN2003 Darlington driver IC between PORTB and the motor coils amplifies the current. The step rate (delay between patterns) controls motor speed.

**Q7. What is the difference between USART and MSSP in PIC16F873?**

**USART:** Handles asynchronous serial (UART) and synchronous serial (SPI) communication on RC6/RC7. **MSSP:** A second serial module supporting both SPI (master/slave) and I2C master/slave modes on RC3/RC4/RC5. Both can be used simultaneously.

**Q8. How is LCD interfaced with PIC16F873 in 4-bit mode?**

Only 4 data lines (D4–D7) are connected to PORTB, along with RS and EN control lines (RW tied to GND). Data is sent as two 4-bit nibbles (high nibble first, then low nibble), each latched by pulsing EN high then low.

**Q9. How does the Watchdog Timer work in PIC16F873?**

The WDT has its own RC oscillator ( $\approx 18$  ms period). Software must execute `CLRWDT` instruction before timeout. If not cleared in time (e.g., due to program hang), WDT causes a device reset. Timeout can be extended using a postscaler (up to  $\times 128 \approx 2.3$  s).

## Frequently Asked / Viva Questions (contd.)

**Q10. What are the Timer modes available in PIC16F873?**

**Timer0:** 8-bit timer/counter with prescaler (1:2 to 1:256). **Timer1:** 16-bit timer/counter, can use external crystal for low-power RTC. **Timer2:** 8-bit timer with prescaler and postscaler; period register PR2 sets overflow point; used for PWM generation.

**Q11. Describe the design of an automated vending machine using PIC16F873.**

- IR sensor (ADC) detects coins; balance tracked in SRAM variable.
- LCD (4-bit mode) shows balance and prompts.
- Keypad selects product; price compared with balance.
- If balance  $\geq$  price: stepper motor (via ULN2003) dispenses product; buzzer beeps.
- Balance decremented by price; “Thank You” displayed.

**Q12. Why is PIC preferred for cost-sensitive embedded applications?**

PIC MCUs are very inexpensive (PIC16F84A ~ Rs. 80–160), available in small packages, have wide voltage range (2–5.5 V), abundant development tools (MPLAB IDE, free compiler), large community, and integrate all needed peripherals on-chip (eliminating external ICs). They are ideal for high-volume, cost-sensitive consumer products.

## PIC16F873 vs. 8051 vs. ARM Cortex-M0 — Comparison

| Feature         | PIC16F873            | 8051 (AT89S52)       | ARM Cortex-M0         |
|-----------------|----------------------|----------------------|-----------------------|
| Architecture    | Harvard RISC (8-bit) | Harvard CISC (8-bit) | Harvard RISC (32-bit) |
| Instructions    | 35 (14-bit words)    | 255 (variable)       | Thumb-2 (16/32-bit)   |
| Max Clock       | 20 MHz               | 33 MHz               | 48 MHz                |
| Flash           | 8 KB                 | 8 KB (ISP)           | 32–128 KB             |
| RAM             | 192 bytes            | 256 bytes            | 8–32 KB               |
| Built-in ADC    | 10-bit, 5-ch         | None                 | 12-bit, 8+ ch         |
| Price (approx.) | Rs. 40–160           | Rs. 65–160           | Rs. 80–400            |
| IDE             | MPLAB X              | Keil, SDCC           | Keil, GCC ARM         |